Ashish Behera

Mechanical Engineering

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Professional Summary

Mechanical Engineering graduate (Bachelor of Technology) with hands-on technical experience as an Apprentice in the Inspection and Quality Department at Hindustan Aeronautics Limited (HAL). Gained comprehensive exposure to KRD-99B engine health monitoring, engine sensor assembly, AL-31FP engine system testing, and hydraulic system assembly and testing. Aspiring to build a career in mechanical design, aerospace quality inspection, intelligent automation, and safety-focused engineering solutions. Proven academic innovation through projects including a smart motorized scissor jack and an AI-based autonomous forklift concept featuring real-time monitoring and web-based control.

Professional Experience

Hindustan Aeronautics Limited (HAL) Koraput, Odisha
Apprentice – Junior Engineer *July 2025 – July 2026*

- Assisted with KRD-99B engine health-related activities and the assembly of various engine sensors.
- Assisted with AL-31FP engine system testing and related inspection activities.
- Supported hydraulic system assembly and testing as part of practical aerospace engineering work.
- Gained hands-on exposure to aerospace component handling, assembly procedures, system testing, and inspection and quality practices.

Academic Projects

Smart Motorized Scissor Jack with Safety Monitoring
Parala Maharaja Engineering College

- Designed a motorized scissor jack concept integrating an electric lifting mechanism with load monitoring and an additional safety-warning approach.
- Incorporated a fallback warning concept intended to alert the user when the jack approaches unsafe operating conditions or potential system failure.
- Designed an audible alert mechanism using beep warnings to indicate abnormal load conditions, motor loading, or potential operational problems.
- Proposed a web-based user interface to communicate faults and abnormal operating conditions to the user for improved monitoring and troubleshooting.
- Focused on lifting safety, load awareness, fault notification, and convenient system monitoring.

AI-Based Autonomous Forklift Machine
Uma Charan Patnaik Engineering School

- Designed an autonomous forklift concept for automated material handling and warehouse transportation operations.
- Incorporated AI-based image processing, object recognition, and object detection concepts to support environmental awareness and navigation.
- Included automatic path detection and shortest-path planning concepts to support efficient movement between warehouse locations.
- Proposed a web-based control interface for forklift operation and warehouse storage and transport management.
- Incorporated battery-status and payload monitoring concepts to help track power availability and lifting-load conditions.
- Included automatic braking and emergency-stop concepts to improve operational safety and help reduce collision risks.

Technical Skills

CAD and Mechanical Design: SolidWorks, AutoCAD, CATIA

Engineering Analysis: ANSYS, Engineering Mechanics, Tolerance Analysis

Aerospace and Quality: Component Inspection, Engine Sensor Assembly, System Assembly and Testing, Quality Control

Mechanical Systems: Hydraulic Systems, Motorized Lifting Mechanisms, Load Monitoring Concepts

Automation and Computer Vision: AI-Based Image Processing, Object Detection, Path Planning, Autonomous Material Handling Concepts

Monitoring and Control: Battery Monitoring, Load Monitoring, Automatic Braking Concepts, Web-Based Control Interfaces

Software and Productivity: IFS ERP System, Microsoft Office, LibreOffice

Professional Skills: Teamwork, Leadership, Problem-Solving, Time Management

Education

Parala Maharaja Engineering College — Berhampur, Odisha, India
B.Tech. in Mechanical Engineering; GPA: 7.35 — *November 2021 – June 2024*

Uma Charan Patnaik Engineering School — Berhampur, Odisha, India
Diploma in Mechanical Engineering; Percentage: 71.60% — *July 2018 – August 2021*

Certifications

The Complete C Programming Course for Beginners: By Md. A. Barik | Udemy.com 2022

Java Complete Course For Begineers 2022: Global Code Institute | Udemy.com 2022

C++ Programming Step By Step From Beginner To Ultimate Level Instructor: Lernen Hub Academy | Udemy.com 2022

Python Programming Complete Beginners Course Bootcamp: By Dmitry Yedunov | Udemy.com 2022

Hobbies & Interests

Programming: Exploring Software Development, Coding, and Problem-Solving

Embedded Systems: Building Embedded Projects and Exploring Microcontrollers

Electronics Projects: Designing and Experimenting with Electronic Circuits and Hardware

Android Development: Exploring Android App Development and Mobile Technologies